

II-Probabilistic Approaches Likelihoods, empirical distributions and learning

Foundations of Neuro-Symbolic AI

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June 22, 2026

Outline

- ▶ Log-likelihood
- ▶ Empirical Distributions
- ▶ Learning Bayesian networks

How to select a model given data?

Let us consider the typical **learning problem**:

- ▶ The distribution $\mathbb{P} [X_{[d]}]$ of variables $X_{[d]}$ is unknown
- ▶ We observe m samples

$$x_{[d]}^j = (x_0^j, \dots, x_{d-1}^j)$$

enumerated by $j \in [m]$.

We assume that each sample $x_{[d]}^j$ is drawn

- ▶ **i**ndependently from each other and
- ▶ **i**dentically **d**istributed by a distribution $\mathbb{P} [X_{[d]}]$.

This assumption is abbreviated by **i.i.d.**.

The probability of drawing a dataset $X_{[d]}^{[m]}$ from a distribution $\mathbb{P} [X_{[d]}]$ is then

$$\prod_{j \in [m]} \mathbb{P} [X_{[d]} = x_{[d]}^j] .$$

This is the **likelihood** of $\mathbb{P} [X_{[d]}]$.

The likelihood principle

We make the hypothesis that the distribution is from a set Γ of possible distributions. A natural choice would be to choose the distribution maximizing the likelihood

$$\operatorname{argmax}_{\mathbb{P}_{[X_{[d]}] \in \Gamma}} \left(\prod_{j \in [m]} \mathbb{P} \left[X_{[d]} = x_{[d]}^j \right] \right) .$$

Sometimes we **regularize**, that is prefer simple distributions over more complex ones.

Likelihood principle

The selection of a distribution should depend on the data $X_{[d]}^{[m]}$ only through the likelihood

$$\prod_{j \in [m]} \mathbb{P} \left[X_{[d]} = x_{[d]}^j \right] .$$

Negative log-likelihood

We simplify the likelihood by taking the log and averaging. This is the **log-likelihood**

$$\frac{1}{m} \ln \left[\prod_{j \in [m]} \mathbb{P} \left[x_{[d]}^j \right] \right] = \frac{1}{m} \sum_{j \in [m]} \ln \left[\mathbb{P} \left[x_{[d]}^j \right] \right] .$$

The maximum likelihood method corresponds then with the minimization of the **negative log-likelihood (NLL)**

$$\operatorname{argmin}_{\mathbb{P}[X_{[d]}] \in \Gamma} \sum_{j \in [m]} - \ln \left[\mathbb{P} \left[X_{[d]} = x_{[d]}^j \right] \right] .$$

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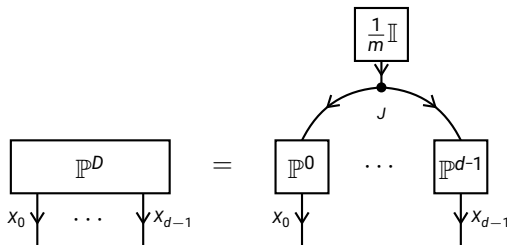
Empirical distributions

Let us understand the uniform choice of a sample from $x_{[d]}^{[m]}$ as a distribution

$$\mathbb{P}^D [X_{[d]}] = \frac{1}{m} \sum_{j \in [m]} \epsilon_{x_{[d]}^j} [X_{[d]}]$$

which we call **empirical distribution**.

We have the CP decomposition of the empirical distribution



where

$$\mathbb{P}^k [X_k | J] = \sum_{j \in [m]} \epsilon_{x_k^j} [X_k] \otimes \epsilon_j [J] .$$

Perspective of basis calculus

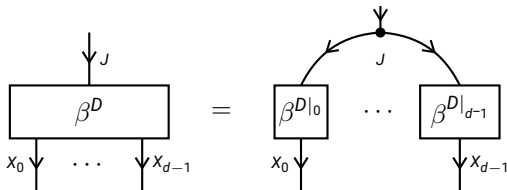
We understand the empirical distribution to samples $x_{[d]}^{[m]}$ as the basis encoding of the sample selecting map

$$D : [m] \rightarrow \bigtimes_{k \in [d]} [m_k]$$

defined for $j \in [m]$ by $D(j) = x_{[d]}^j$. To be more precise we have

$$\mathbb{P}^D [X_{[d]}] = \left\langle \beta^D [X_{[d]}, J], \frac{1}{m} \mathbb{I} [J] \right\rangle_{[X_{[d]}]}.$$

The CP decomposition of the empirical distribution is then based on the decomposition of the basis encoding into the basis encodings of its coordinate restrictions:



Cross-entropy

The **cross-entropy** between two distributions $\tilde{\mathbb{P}}[X_{[d]}]$ and $\mathbb{P}[X_{[d]}]$ is

$$\mathbb{H}[\tilde{\mathbb{P}}, \mathbb{P}] = \left\langle \tilde{\mathbb{P}}[X_{[d]}], -\ln[\mathbb{P}[X_{[d]}]] \right\rangle_{[\emptyset]}.$$

The negative log-likelihood is the cross-entropy between the empirical distribution and the candidate $\mathbb{P}[X_{[d]}]$.

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Two learning tasks

- ▶ Structure learning: Find the hypergraph $\mathcal{G}$
- ▶ Parameter estimation: Find the best decorations by conditional probability distributions

Decomposing the negative log-likelihood

For a Bayesian network we have for any index tuple $x_{[d]}$

$$\ln [\mathbb{P} [X_{[d]} = x_{[d]}]] = \sum_{k \in [d]} \ln [\mathbb{P} [X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] .$$

The NLL thus decomposes into local terms

$$\begin{aligned} & \left\langle \mathbb{P}^D [X_{[d]}] , -\ln [\mathbb{P} [X_{[d]}]] \right\rangle_{[\emptyset]} \\ &= \left\langle \mathbb{P}^D [X_{[d]}] , \sum_{k \in [d]} -\ln [\mathbb{P} [X_k | X_{\text{Pa}(k)}]] \right\rangle_{[\emptyset]} \\ &= \sum_{k \in [d]} \left(\left\langle \mathbb{P}^D [X_{[d]}] , -\ln [\mathbb{P} [X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] \right\rangle_{[\emptyset]} \right) \end{aligned}$$

The minimum can found separately for each k by the minimization problem

$$\operatorname{argmin}_{\mathbb{P} [X_k | X_{\text{Pa}(k)}]} \frac{1}{m} \sum_{j \in [m]} -\ln [\mathbb{P} [X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] .$$

Optimum by normalization of the empirical distribution

The optima for each conditional distribution is taken at

$$\begin{aligned} \operatorname{argmin}_{\mathbb{P}[X_k | X_{\text{Pa}(k)}]} & \left\langle \mathbb{P}^D, -\ln [\mathbb{P}[X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] \right\rangle_{[\emptyset]} \\ &= \left\langle \mathbb{P}^D [X_{[d]}] \right\rangle_{[X_k | X_{\text{Pa}(k)}]} \end{aligned}$$

This can be verified by the first order criterion (vanishing gradients).

Parameter estimation

Parameter estimation for Bayesian networks

- ▶ The likelihood of Bayesian networks decomposes into local terms.
- ▶ Best log-likelihood maximizers are the conditional probability distributions.